P2 Chapter 9: Differentiation

Parametric Differentiation

Parametric Differentiation

Recall from the previous chapter that parametric equations are when we define each of x and y (and possibly z) in terms of some separate parameter, e.g. t.

✏ If x and y are given as functions of a parameter t, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Proof: By Chain Rule: $\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$

Then rearrange.

[Textbook] Find the gradient at the point P where $t = 2$, on the curve given parametrically by

$x = t^3 + t, \qquad y = t^2 + 1, \qquad t \in \mathbb{R}$

$$\frac{dy}{dt} = 2t \qquad \frac{dx}{dt} = 3t^2 + 1$$

$$\therefore \frac{dy}{dx} = \frac{2t}{3t^2 + 1}$$

When $t = 2, \frac{dy}{dx} = \frac{4}{13}$

[Textbook] Find the equation of the normal at the point P where $\theta = \frac{\pi}{6}$, to the curve with parametric equations $x = 3\sin\theta$, $y = 5\cos\theta$

$$\frac{dx}{d\theta} = 3\cos\theta, \frac{dy}{d\theta} = -5\sin\theta \qquad 0 \le \theta < 2\pi$$

$$\therefore \frac{dy}{dx} = -\frac{5\sin\theta}{3\cos\theta} = -\frac{5}{3}\tan\theta$$

When $\theta = \frac{\pi}{6}$,

$\frac{dy}{dx} = -\frac{5}{3\sqrt{3}}$, $y = 5\cos\frac{\pi}{6} = \frac{5\sqrt{3}}{2}$, $x = \frac{3}{2}$

$\therefore$ **Equation of normal:**

$$y - \frac{5\sqrt{3}}{2} = \frac{3\sqrt{3}}{5}\left(x - \frac{3}{2}\right)$$

Test Your Understanding

C4 June 2012 Q6

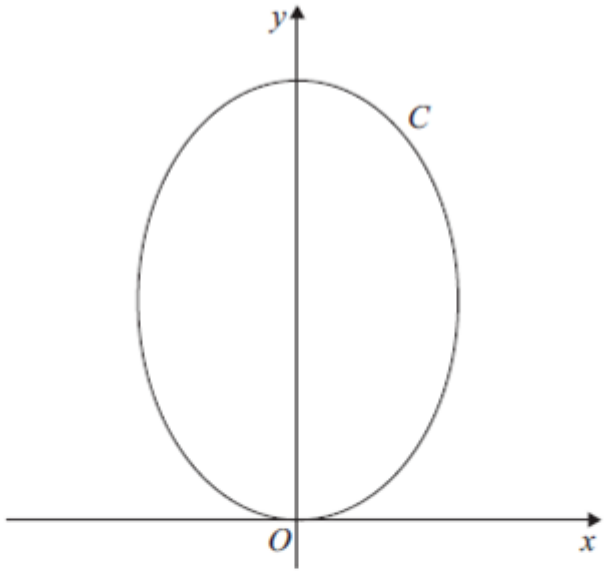

Figure 2

Figure 2 shows a sketch of the curve C with parametric equations

$$x = \sqrt{3}\sin 2t, \qquad y = 4\cos^2 t, \qquad 0 \le t \le \pi.$$

(a) Show that $\frac{dy}{dx} = k\sqrt{3}\tan 2t$, where k is a constant to be determined.

(5)

(b) Find an equation of the tangent to C at the point where $t = \frac{\pi}{3}$.

Give your answer in the form $y = ax + b$, where a and b are constants.

(4)

(a)

?

(b)

?

Test Your Understanding

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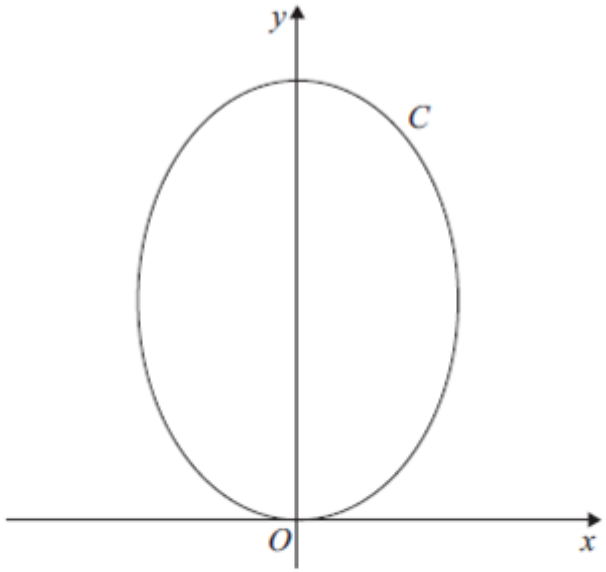

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(4)

(a)	$\frac{dx}{dt} = 2\sqrt{3}\cos 2t$	B1
	$\frac{dy}{dt} = -8\cos t \sin t$	M1 A1
	$\frac{dy}{dx} = \frac{-8\cos t \sin t}{2\sqrt{3}\cos 2t}$	M1
	$= -\frac{4\sin 2t}{2\sqrt{3}\cos 2t}$	
	$\frac{dy}{dx} = -\frac{2}{3}\sqrt{3}\tan 2t \qquad \left(k = -\frac{2}{3}\right)$	A1
(b)	When $t = \frac{\pi}{3}$ $\quad x = \frac{3}{2},\ y = 1$ can be implied	B1
	$m = -\frac{2}{3}\sqrt{3}\tan\left(\frac{2\pi}{3}\right) \quad (= 2)$	M1
	$y - 1 = 2\left(x - \frac{3}{2}\right)$	M1
	$y = 2x - 2$	A1

Exercise 9.7

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Homework Exercise

1 Find $\frac{dy}{dx}$ for each of the following, leaving your answer in terms of the parameter t.

a $x = 2t,\ y = t^2 - 3t + 2$ **b** $x = 3t^2,\ y = 2t^3$ **c** $x = t + 3t^2,\ y = 4t$

d $x = t^2 - 2,\ y = 3t^5$ **e** $x = \frac{2}{t},\ y = 3t^2 - 2$ **f** $x = \frac{1}{2t-1},\ y = \frac{t^2}{2t-1}$

g $x = \frac{2t}{1+t^2},\ y = \frac{1-t^2}{1+t^2}$ **h** $x = t^2e^t,\ y = 2t$ **i** $x = 4\sin 3t,\ y = 3\cos 3t$

j $x = 2 + \sin t,\ y = 3 - 4\cos t$ **k** $x = \sec t,\ y = \tan t$ **l** $x = 2t - \sin 2t,\ y = 1 - \cos 2t$

m $x = e^t - 5,\ y = \ln t,\ t > 0$ **n** $x = \ln t,\ y = t^2 - 64,\ t > 0$ **o** $x = e^{2t} + 1,\ y = 2e^t - 1,\ -1 < t < 1$

2 **a** Find the equation of the tangent to the curve with parametric equations $x = 3 - 2\sin t$, $y = t\cos t$, at the point P, where $t = \pi$.

b Find the equation of the tangent to the curve with parametric equations $x = 9 - t^2$, $y = t^2 + 6t$, at the point P, where $t = 2$.

3 **a** Find the equation of the normal to the curve with parametric equations $x = e^t$, $y = e^t + e^{-t}$, at the point P, where $t = 0$.

b Find the equation of the normal to the curve with parametric equations $x = 1 - \cos 2t$, $y = \sin 2t$, at the point P, where $t = \frac{\pi}{6}$

4 Find the points of zero gradient on the curve with parametric equations

$$x = \frac{t}{1-t},\quad y = \frac{t^2}{1-t},\quad t \neq 1$$

You do not need to establish whether they are maximum or minimum points.

Homework Exercise

5 The curve C has parametric equations $x = e^{2t}$, $y = e^t - 1$, $t \in \mathbb{R}$.

a Find the equation of the tangent to C at the point A where $t = \ln 2$.

b Show that the curve C has no stationary points.

6 The curve C has parametric equations

$$x = \frac{t^2 - 3t - 4}{t}, \quad y = 2t, \quad t > 0$$

The line l_1 is a tangent to C and is parallel to the line with equation $y = x + 5$.
Find the equation of l_1. **(8 marks)**

7 A curve has parametric equations

$$x = 2\sin^2 t, \quad y = 2\cot t, \quad 0 < t < \frac{\pi}{2}$$

a Find an expression for $\frac{dy}{dx}$ in terms of the parameter t. **(4 marks)**

b Find an equation of the tangent to the curve at the point where $t = \frac{\pi}{6}$ **(4 marks)**

8 The curve C has parametric equations

$$x = 4\sin t, \qquad y = 2\operatorname{cosec} 2t, \qquad 0 \leqslant t \leqslant \pi$$

The point A lies on C and has coordinates $\left(2\sqrt{3}, \frac{4\sqrt{3}}{3}\right)$.

a Find the value of t at the point A. **(2 marks)**

The line l is a normal to C at A.

b Show that an equation for l is $9x + 12y - 34\sqrt{3} = 0$. **(6 marks)**

Homework Exercise

9 The curve C has parametric equations

$$x = t^2 + t, \quad y = t^2 - 10t + 5, \quad t \in \mathbb{R}$$

where t is a parameter. Given that at point P, the gradient of C is 2,

a find the coordinates of P **(4 marks)**

b find the equation of the tangent to C at point P **(3 marks)**

c show that the tangent to C at point P does not intersect the curve again. **(5 marks)**

Problem-solving

Substitute the equations for x and y into the equation of your tangent, and show that the resulting quadratic equation has one unique root.

10 The curve C has parametric equations

$$x = 2\sin t, \quad y = \sqrt{2}\cos 2t, \quad 0 < t < \pi$$

a Find an expression for $\frac{dy}{dx}$ in terms of t. **(2 marks)**

The point A lies on C where $t = \frac{\pi}{3}$. The line l is the normal to C at A.

b Find an equation for l in the form $ax + by + c = 0$, where a, b and c are exact constants to be found. **(5 marks)**

c Prove that the line l does not intersect the curve anywhere other than at point A. **(6 marks)**

Homework Exercise

11 A curve has parametric equations

$$x = \cos t, \quad y = \tfrac{1}{2}\sin 2t, \quad 0 \leqslant t < 2\pi$$

a Find an expression for $\frac{dy}{dx}$ in terms of t. **(2 marks)**

b Find an equation of the tangent to the curve at point A where $t = \frac{\pi}{6}$ **(4 marks)**

The lines l_1 and l_2 are two further distinct tangents to the curve. Given that l_1 and l_2 are both parallel to the tangent to the curve at point A,

c find an equation of l_1 and an equation of l_2 **(6 marks)**

Homework Answers

1 **a** $\frac{2t-3}{2}$ **b** $\frac{6t^2}{6t} = t$ **c** $\frac{4}{1+6t}$ **d** $\frac{15t^3}{2}$

e $-3t^3$ **f** $t(1-t)$ **g** $\frac{2t}{t^2-1}$ **h** $\frac{2}{(t^2+2t)e^t}$

i $-\frac{3}{4}\tan 3t$ **j** $4\tan t$ **k** $\operatorname{cosec} t$ **l** $\cot t$

m $\frac{1}{te^t}$ **n** $2t^2$ **o** $\frac{1}{e^t}$

2 **a** $y = -\frac{1}{2}x + \frac{3}{2} - \pi$ **b** $2y + 5x = 57$

3 **a** $x = 1$ **b** $y + \sqrt{3}x = \sqrt{3}$

4 (0, 0) and (−2, −4)

5 **a** $y = \frac{1}{4}x$

b $\frac{dy}{dx} = \frac{1}{2}e^{-t} = 0 \Rightarrow e^{-t} = 0$

No solution, therefore no stationary points.

6 $y = x + 7$

7 **a** $-\frac{1}{2}\sec t \operatorname{cosec}^3 t$ **b** $8x + \sqrt{3}y - 10 = 0$

8 **a** $\frac{\pi}{3}$

b $\frac{dy}{dt} = -4\cot 2t \operatorname{cosec} 2t, \frac{dx}{dt} = 4\cos t$

$\frac{dy}{dx} = \frac{-4\cot 2t \operatorname{cosec} 2t}{4\cos t} = \frac{-\cot 2t \operatorname{cosec} 2t}{\cos t}$

At $t = \frac{\pi}{3}, \frac{dy}{dx} = \frac{4}{3}$

Gradient of normal: $-\frac{3}{4}$

Equation of normal:

$y - \frac{4\sqrt{3}}{3} = -\frac{3}{4}(x - 2\sqrt{3}) \Rightarrow 9x + 12y - 34\sqrt{3} = 0$

9 **a** (30, 101) **b** $y = 2x + 41$

c $t^2 - 10t + 5 = 2(t^2 + t) + 41$

$t^2 - 10t + 5 = 2t^2 + 2t + 41$

$0 = t^2 + 12t + 36$

Discriminant $= 12^2 - 4 \times 1 \times 36 = 144 - 144 = 0$

Therefore the curve and the line only intersects once.

Therefore it does not intersect the curve again.

10 **a** $-2\sqrt{2}\sin t$ **b** $x - \sqrt{6}y - 2\sqrt{3} = 0$

c $2\sin t - \sqrt{12}\cos 2t - 2\sqrt{3} = 0$

$\sin t - \sqrt{3}\cos 2t - \sqrt{3} = 0$

$2\sqrt{3}\sin^2 t + \sin t - 2\sqrt{3} = 0$

$(2\sin t - \sqrt{3})(\sqrt{3}\sin t + 2) = 0$

$\sin t = \frac{\sqrt{3}}{2}\left(\text{sint } t \neq -\frac{2}{\sqrt{3}}\right) \Rightarrow t = \frac{\pi}{3} \text{ or } \frac{2\pi}{3}$

B is when $t = \frac{2\pi}{3}$: $\left(2\sin\frac{2\pi}{3}, \sqrt{2}\cos\frac{4\pi}{3}\right) = \left(\sqrt{3}, -\frac{1}{\sqrt{2}}\right)$

Same point as A, so l only intersects C once.

11 **a** $-\frac{\cos 2t}{\sin t}$ **b** $y = -x + \frac{3\sqrt{3}}{4}$

c $y = -x$ and $y = -x - \frac{3\sqrt{3}}{4}$